

题目

已知 H_2S 的 $K_{\text{a}1} = 1.1 \times 10^{-7}$, $K_{\text{a}2} = 1.3 \times 10^{-13}$; $K_{\text{sp}}(\text{CuS}) = 6.3 \times 10^{-36}$, $K_{\text{sp}}(\text{Cu}_2\text{S}) = 2.5 \times 10^{-48}$ 。 $\varphi_{\text{A}}^{\theta}(\text{S}/\text{H}_2\text{S}) = 0.142 \text{ V}$, $\varphi_{\text{B}}^{\theta}(\text{S}/\text{S}^{2-}) = -0.476 \text{ V}$; $\varphi_{\text{A}}^{\theta}(\text{NO}_3^-/\text{NO}) = 0.942 \text{ V}$, $\varphi^{\theta}(\text{Cu}^{2+}/\text{Cu}) = 0.342 \text{ V}$, $\varphi^{\theta}(\text{Cu}^+/\text{Cu}) = 0.521 \text{ V}$ 。

根据以上条件, 写出 Cu_2S 溶于稀硝酸生成蓝色溶液与黄色沉淀的离子方程式, 利用Hess 定律等计算该反应的标准平衡常数。选择最接近你计算的平衡常数的选项。

A. 其他选项均不正确

B. $K^{\theta} = 1.1 \times 10^{80}$

C. $K^{\theta} = 2.0 \times 10^{130}$

D. $K^{\theta} = 2.4 \times 10^8$

E. $K^{\theta} = 5.1 \times 10^{43}$

答案

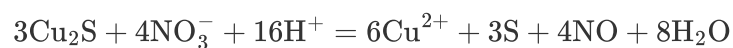
正确答案: **B**

详细解析

首先计算二价铜还原到一价铜的标准电势:

$$\varphi_B^\theta(\text{Cu}^{2+}/\text{Cu}^+) = 2\varphi^\theta(\text{Cu}^{2+}/\text{Cu}) - \varphi^\theta(\text{Cu}^+/\text{Cu}) = 0.163 \text{ V}$$

题中所述化学方程式为:



CHECKPOINT

1 PTS

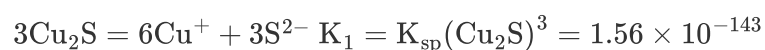
化学方程式为: $3\text{Cu}_2\text{S} + 4\text{NO}_3^- + 16\text{H}^+ = 6\text{Cu}^{2+} + 3\text{S} + 4\text{NO} + 8\text{H}_2\text{O}$

将该反应进行分解, 使得沉淀溶解平衡和氧化还原反应被分开:

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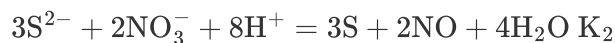
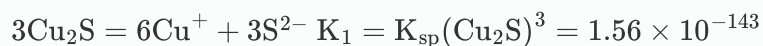
1 PTS

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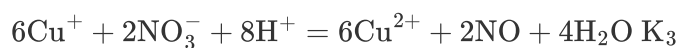
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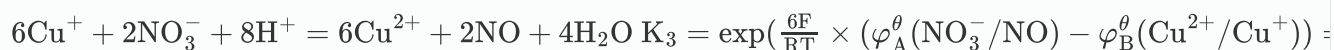
CHECKPOINT

2 PTS



CHECKPOINT

2 PTS



其中，对于 K_2 和 K_3 ，有

$$K_2 = \exp\left(\frac{6F}{RT} \times (\varphi_A^\theta(\text{NO}_3^-/\text{NO}) - \varphi_B^\theta(\text{S}/\text{S}^{2-}))\right) = 6.64 \times 10^{143}$$

$$K_3 = \exp\left(\frac{6F}{RT} \times (\varphi_A^\theta(\text{NO}_3^-/\text{NO}) - \varphi_B^\theta(\text{Cu}^{2+}/\text{Cu}^+))\right) = 1.03 \times 10^{79}$$

$$K^\theta = K_1 K_2 K_3 = 1.1 \times 10^{80}$$

CHECKPOINT

1 PTS

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